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13 APRIL 1955
SUPERSEDING
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7 MAY 1953

MILITARY SPECIFICATION

# TEST SET, TELEPHONE (TEST SET TS–190( )/U)

*This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy and the Air Force*

## 1. SCOPE

1.1 Test Set TS–190( )/U consists of a 1000 ohm telephone receiver with a switch controlled resistance of 50,000 ohms It is used to locate faults such as opens, grounds, shorts, and crosses and to trace continuity of circuits on telephone and telegraph equipment When a circuit is completed through the test set between negative battery and ground, a click is heard in the receiver When tracing circuits the test set is used to listen for tones

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids form a part of this specification

### SPECIFICATIONS

FEDERAL

NN–B–591 —Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment)
NN–B–621 —Boxes, Wood, Nailed and Lock-Corner.
NN–B–631 —Boxes, Wood, Wirebound (For Domestic Shipment).
QQ–S–571 —Solder; soft (Tin, Tin-Lead, and Lead Silver).
UU–T–111 —Tape, Paper, Gummed (Sealing and Securing)
LLL–B–631 —Boxes, Fiber, Corrugrated (For Domestic Shipment)
PPP–B–601 —Boxes, Wood Cleated-Plywood
PPP–B–676 —Boxes, Set-Up, Paperboard

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JAN–P–106 —Packaging and Packing For Overseas Shipment—Boxes, Wood, Mailed
MIL–B–107 —Boxes, Wood, Wirebound (Overseas Type)
JAN–P–108 —Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board) Exterior and Interior
MIL–P–116 —Preservation, Methods of
JAN–P–120 —Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard
JAN–P–127 —Packaging and Packing For Overseas Shipment — Tape, Adhesive, Pressure-Sensitive, Water Resistant.
MIL–B–131 —Barrier - material, water vaporproof, flexible.
JAN–T–152 —Treatment, Moisture and Fungus—Resistant of Communica-

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tions, Electronic and Associated Electrical Equipment; general process for.
MIL-C-3849 —Cord, electrical (tinsel).
MIL-L-10547 —Liners, case, waterproof.
MIL-P-11268 —Parts, materials, and Processes used in Signal Corps Equipment

2.2 **Other publications.** — The following documents form a part of this specification Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

Association of American Railroads
Uniform Freight Classification Ratings, Rules and Regulations

(Copies of Uniform Freight Classification, Ratings, Rules and Regulations may be obtained from the Association of American Railroads, 59 East Van Burer Street Chicago, Illinois.)

National Motor Freight Association
National Motor Freight Classification Rules and Container Specifications for Truck Shipments.

(Copies of Classification Rules and Container Specifications for Truck Shipments may be obtained from the National Motor Freight Association, 1424 - 16th Street, N W, Washington 6, D C )

U. S ARMY
72-53 —Finishes (For Ground Signal Equipment)

## STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-169 —Extreme Temperature Cycle.
MIL-STD-170 —Moisture - Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252 —Wired Equipment, Classification of Visual and Mechanical Defects For

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer Both the title and identifying number or symbol should be stipulated when requesting copies )

## 3. REQUIREMENTS

3.1 **Procurement model.**—The procurement model shall be the one which will be available for inspection by prospective bidders and will be lent to the contractor, in accordance with the invitation for bids Unless otherwise specified herein or in the invitation for bids, the equipment shall conform to the following

3.1.1 *Construction* —Physical construction of the equipment shall conform to the procurement model

3.1.2 *Features* —The equipment shall incorporate all features of the procurement model

3.1.3 *Verification* —As soon as the procurement model is received, and with the minimum necessary adjustment and alinement (if any), the contractor shall measure its performance for compliance with specified characteristics of the equipment on contract If the model does not meet the specified performance characteristics, the contractor shall adjust or aline it for optimum performance and then repeat the measurements The contractor shall immediately prepare copies of a test report on the measurements, and forward them as required by the contract In case of conflict between specified performance characteristics and performance of the model the former shall govern

3.2 **Preproduction samples.**

3.2.1 *Quantities.* — Two complete equipments shall be furnished for Government in-



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spection and approval. These shall be in addition to preproduction samples of parts, materials, and processes specified by Specification MIL-P-11268.

3.2.2 *Compliance with specified requirements.*

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production and shall be accompanied by a statement to that effect (See 3 2 2 3 for exception )

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement After being released to the contractor, preproduction samples to be offered as units on contract shall be refabricated by the contractor, if necessary, to meet specified requirements

3.2.2.3 When deviation from 3 2 2 1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract

3.2.3 *Test data* —The preproduction samples shall be accompanied by test data showing compliance with specified performance The test data shall comprise an engineering report giving test procedure, observations and other data, calculations, test results and essential details of the testing equipment (manufacturer's model serial number, date of calibration, and the like)

3.2.4 *Reference standards* —After preproduction samples have been approved and returned to the contractor's plant, they shall be kept intact in custody of the Government inspector until released by him They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 **Parts, materials and processes.**—Parts, materials, and processes shall be in accordance with the applicable requirements of Specification MIL-P-11268, unless otherwise specified herein.

3.3.1 *Insulation material* —Insulation material which will support rapid combustion shall not be used All insulating material except that used on wire shall be sufficiently rigid and free from heat distortion to maintain its function when the equipment in which it is used is exposed to its maximum specified ambient temperature at 100 percent relative humidity while in operation, storage, or during transportation

3.3.2 *Metals* —The metals used shall be of the proper alloy and hardness necessary to provide the required strength, and resistance to wear Materials shall be of corrosion resisting types, or shall be suitably processed to resist corrosion in accordance with Specification 72-53 Dissimilar metals used in contact with each other shall be carefully selected in accordance with Specification 72-53 to prevent electrolytic corrosion in the presence of moisture

3.3.3 *Tropicalization*

3.3 3.1 *Treatment of assembled equipment* —Assembled equipment shall be treated in accordance with Specification JAN-T-152 unless both of the following conditions are met

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual detailed Government document (specification standard, drawing, etc ) which is applicable on contract

(2) It does not incorporate any material listed in 3 3 2.2 unless the material has been given the treatment specified in Specification MIL-P-11268 or a treatment approved by the contracting officer.



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(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application

(b) The finish of the interior of the equipment meets the requirements of Specification 72–53

3.3.3.2 *Materials not resistant to moisture and fungi*.—The following materials are considered not resistant to moisture and fungi

(a) Cellulose, regenerated

(b) Cotton.

(c) Cork

(d) Felt, hair or wool

(e) Jute

(f) Leather.

(g) Linen

(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.

(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base

3.3.3.3 *Statement of treatment*.—The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use This statement shall include the statements required by Specifications MIL–P–11268 and JAN–T–152, as applicable

3.4 **Ruggedness and reliability.**—All parts and details of construction shall have the maximum of ruggedness and reliability under all operating conditions The equipment shall be so constructed that it will withstand long periods of service and repeated rough handling in transportation, with a minimum of damage or readjustment.

3.5 **Temperature and humidity.** — The equipment shall be capable of satisfactory operation after storage at any temperature within the ranges from −80° to +160°F, at any relative humidity up to 100 percent The equipment shall be capable of operating in any external ambient temperature between the limits of 0° and 132°F., and all ranges of humidity All items or parts entering into the construction of any unit of the complete equipment shall be capable of operating continuously, without damage, at any external ambient temperature in the operating range specified herein

3.6 **Interchangeability.** — All components, and all parts of any component, shall be mechanically and electrically interchangeable with the model, and all other corresponding components or parts on the order

3.7 **Wiring**.—The equipment shall be wired in accordance with the schematic diagram shown on figure 1

3.7.1 *Receiver* — The receiver shall be Western Electric Co type #509, or equal, having a dc resistance of 1100 ohms ± 10 percent A 50,000 ohm ± 10 percent resistor shall be connected in series with the receiver windings which shall be located in the Western Electric Co tool No 518A, or equal

3.7.1.1 *Headband*.—The headband shall be Western Electric Company type #11A or #15A, or equal

3.7.2 *Cord*.—The cord shall be two conductor, tinsel class NS, type 2, natural rubber insulated, Buna S jacketed in accordance with Specification sheet MIL–C–3849/4 with sliding thermosetting plastic sleeves and a length of approximately 72 inches

3.7.3 *Test connectors*.—The free end of one cord shall be equipped with a tool, Western Electric Company type 360A, or equal, also a Western Electric Company #KS–0279 spring clip tool, or equal. A Western Electric Company #518A tool, or equal, with test



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pick and shorting switch which includes a 50,000 ohm resistor installed in a Western Electric Company ±360A tool, or equal, shall be connected to the free end of the other cord.

3.7.4 *Soldering.*

3.7.4.1 *Solder* —Solder used for electrical connections shall be composition Sn60 conforming to Specification QQ-S-571

3.7.4.2 *Flux and cleaning agents* —No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine

3.7.4.3 *Process* —There shall be no sharp points or rough surfaces resulting from insufficient heating The solder shall feather out to a thin edge indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated The minimum necessary amount of flux and solder shall be used for electrical connections Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened

**3.8 Workmanship.**—The equipment shall be manufactured and assembled in a thoroughly workmanlike manner. In particular, soldering shall meet all the requirements of paragraph 3.7.4.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Classification of inspection.**—Inspection shall be classified as follows:

(a) Preproduction inspection

(b) Acceptance inspection

**4.2 Preproduction inspection.**—Preproduction inspection shall consist of the group A inspection and nondestructive group C inspection specified in Tables II and III and the preproduction inspection specified in Table I Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirement

**4.3 Acceptance inspection** —Acceptance inspection shall consist of group A, and group C inspection as specified in 4.3 1 through 4 3 4 Group C inspection shall normally be performed on sample units that have been subjected to and met group A inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group C inspection Each unit which will be subjected to group A, or group C inspection shall be preconditioned in accordance with 4.4

4.3.1 *Group A inspection* —Group A inspection (including sampling) shall conform to Table II and Standard MIL-STD-105 Group A inspection shall be performed in any order which is satisfactory to the Government inspector

TABLE I *Preproduction inspection*

| Inspection | Requirements paragraph | Test paragraph |
|---|---|---|
| Temperature ... ... ... ... | 3.5 | 4.8 |

4.3.2 *Group C inspection.*—Group C inspection shall be as listed in Table III.

4.3.2.1 *Sampling for group C.*—One sample from each 100 units, unless otherwise



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Test Set, Telephone
(Test Set TS-190( )/U) Sheet #6

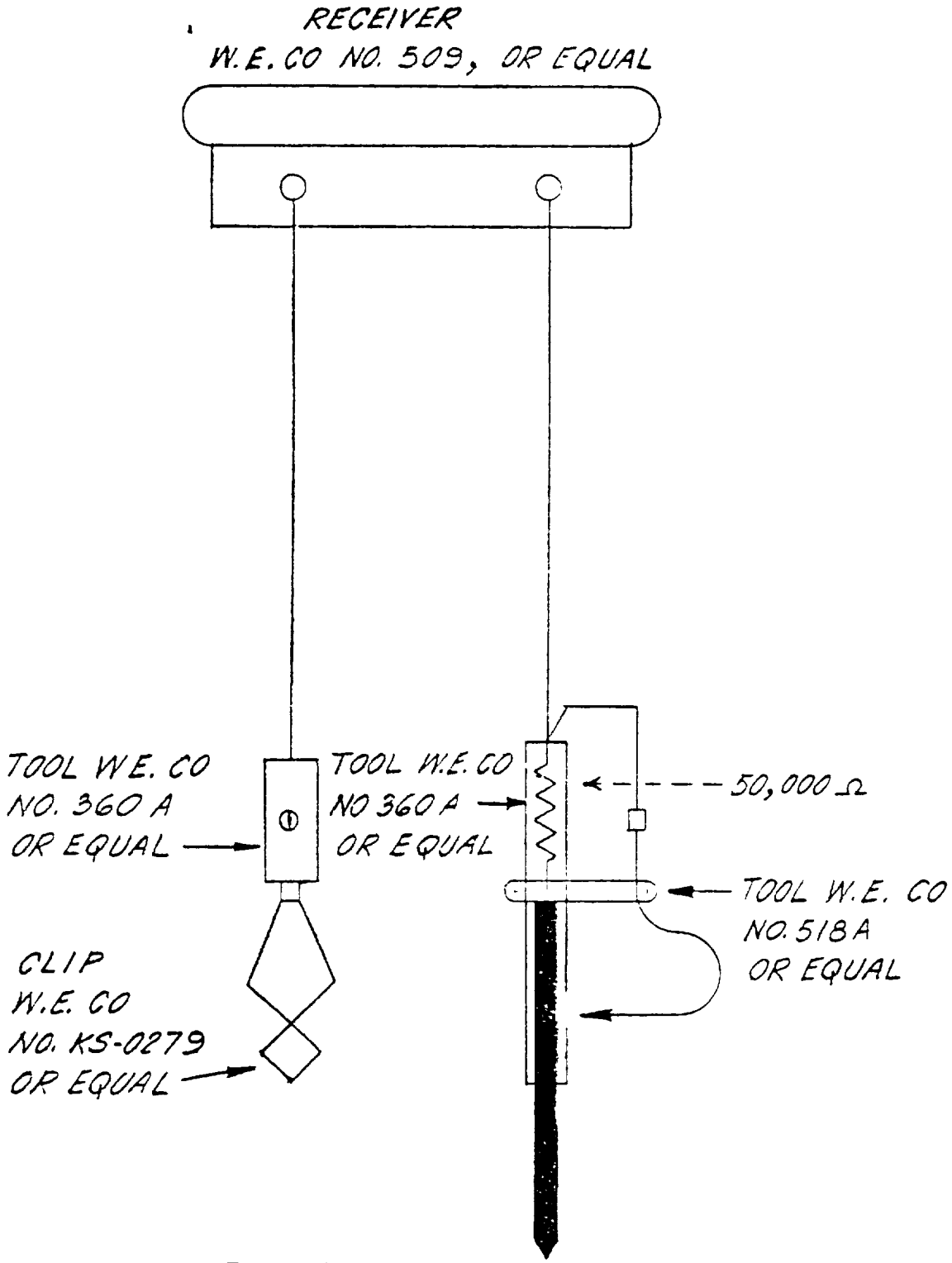



FIGURE 1 *Schematic Diagram for Test Set TS-190( )/U*



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specified, shall be subjected to class C inspection.

TABLE II. *Government group A inspection*

| Inspection | Requirements paragraph | Test paragraph | AQL |
|---|---|---|---|
| Visual and Mechanical, Major | | 4 5 | 1% |
| Visual and Mechanical, Minor | | 4 5 | 3% |
| Continuity | 3 7 | 4 6 1 | 1% |
| Resistance | 3.7 | 4 6 2 | Major for Group |
| Operational | 3 7 | 4 6 3 | |

4.3.2.2 *Noncompliance* —If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and report the results thereof and details of the corrective action taken on the process and affected portion of product to the Government inspector. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE III *Government group C inspection*

| Inspection | Requirements paragraph | Test paragraph |
|---|---|---|
| Moisture Resistance | 3 5 | 4 7 |

4.3.3 *Reinspection of conforming group C sample units.* — Unless otherwise specified sample units which have been subjected to and passed group C inspection, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage

4.3.4 *Disposition of nonconforming product.* — When defective sample units or rejected lots are resubmitted or affected portion of product is submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government

4.4 **Bounce preconditioning.** — After final assembly, each component which will be examined, measured, or tested shall be preconditioned as follows: The component, with shock mounts (if any) removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the L. A B Corporation, Summit, New Jersey, or equal The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces The component shall be subjected to this preconditioning for 1 minute

4.5 **Visual and mechanical inspection.**— Parts and equipment shall be examined for the defects listed in Standard MIL–STD–252.

4.6 **Electrical tests.**

4.6.1 *Continuity test* — Continuity test shall be made on all electrical circuits This test shall be made with the switch in both the normal operating position and the depressed position

4.6.2 *Resistance test* —The resistance of the receiver and the resistance of 50,000 ohm resistor shall be measured using a suitable low potential d.c method for compliance with the respective requirements of 3.7.1

4.6.3 *Operational test* —With Test Set TS–190 ( )/U completely assembled, the switch shall be given an operational test to verify that depressing the button switch on the side of the Western Electric Co. #518A tool (or equal) shorts out the resistor.

4.7 **Moisture-resistance.** — The following test shall be performed on units selected for group C inspection.



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4.7.1 *Test conditions.*

(a) Do not move components from the humidity chamber for measurements

(b) Start measurements not more than 5 minutes after power is applied to the components. Complete measurements as rapidly as possible Do not leave power on after measurements have been completed

4.7.2 *Test procedure* — The components will be tested as follows·

(a) Dry at 130° ± 5°F. for 24 hours

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4 3 readjust or realine as necessary to meet full specification requirements

(d) Subject to continuous cycling for 5 of the 48-hour cycles Temperature, relative humidity, and period of time for each portion of the cycle shall conform to MIL–STD–170 (Sig C) The measurements specified in 4 3 shall be made during the periods shown on the drawing

(e) After cycling has been completed condition the components for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity Then adjust for optimum performance, using only those means provided by the components No repair or replacement of parts shall be made After adjustment, the components shall meet full specification requirements for those measurements specified in 4.3.

**4.8 Temperature.**—The equpiment shall be subjected to the temperature cycle shown on MIL–STD–169(Sig C). Tests shall be made in accordance with 4.6 during steps 4 and 10. There shall be no degradation in performance from that shown prior to the temperature tests.

## 5. PREPARATION FOR DELIVERY

### 5.1 Preservation and packaging.

5.1 1 *Domestic immediate use* —Test Set TS–190( ) /U shall be packaged for shipment in conformance with the requirements of Uniform Freight Classification, Ratings, Rules and Regulations for Rail Shipments, National Motor Freight Classification Rules and Container Specifications for Truck Shipments, Parcel Post Regulations, and the regulations of other carriers as applicable to the mode of transportation

5.1.2 *Domestic Limited storage* —Preserve and package Test Set TS–190( )/U in accordance with procedures specified for the designated method as prescribed in Specification MIL–P–116, and further described in the following paragraphs·

5.1.2.1 *Test Set TS-190( )/U* — Package each test set Method IA–8 as follows Wind cords into a coil of proportionate dimensions Tie in three places with cotton tape, securing plugs and headsets to coil with ties Cushion each test set by wrapping with flexible corrugated paper Secure the wrapping with gummed paper tape, conforming to Specification UU–T–111 Inclose each cushioned item within a close fitting bag fabricated of water–vapor proof barrier material conforming to the requirements of MIL–B–131, class 2

5.1.2.2 *Intermediate package.* — Place a quantity of test sets, packaged as specified in 5 1 2 1 bearing the same stock number, within a close fitting folding carton or set up box conforming to Specification JAN–P–120 or PPP–B–676, respectively. The gross weight of intermediate packages shall not exceed 5 pounds. Fill all voids with flexible corrugated paper to prevent movement. Seal



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box closure with gummed paper tape, conforming to Specification UU-T-111.

5.1.3 *Military overseas* — Preserve and package Test Set TS-190( )/U as described in 5 1.2

5.2 **Packing.**

5.2.1 *Domestic, immediate use* —Test Set TS-190( )/U shall be packed for shipment in conformance with the requirements of Uniform Freight Classification Rules and Container Specifications for Rail Shipment, National Motor Freight Classification Rules and Container Specifications for Truck Shipments, Parcel Post Regulations or other carrier regulations as applicable to mode of transportation

5.2.2 *Domestic, limited storage* —Test Set TS-190( )/U shall be packed in wood cleated fiberboard, wood cleated plywood, nailed wood, corrugated fiberboard or wirebound boxes conforming to Specifications NN-B-591, PPP-B-601, NN-B-621, LLL-B-631 or NN-B-631 The shipping container shall fit the contents snugly Gross weight shall not exceed 65 pounds for fiberboard boxes and 200 pounds for plywood or wood boxes Closure shall be made in accordance with the applicable box specification

5.2.3 *Military overseas* —Test Set TS-190 ( )/U packaged as specified in 5 1 3 shall be packed in wood-cleated, plywood, nailed wood or wirebound boxes conforming to the requirements of Specification PPP-B-601, JAN-P-106 or MIL-B-107 respectively Fabricate the box to fit the contents snugly Unless otherwise specified, shipping containers shall be lined with a sealed waterproof case liner conforming to Specification MIL-L-10547. The approximate gross weight of any exterior shipping container shall not exceed 150 pounds. Close the shipping container as specified in the applicable box specification.

5.2.3.1 Shipping containers shall be strapped in conformance with the appendix of the applicable container specification *only* for direct shipment to ports

5.3 **Marking.**

5.3.1 *Marking for shipment* — Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129

6. **NOTES**

6.1 **Intended use.**—Test Set TS-190( )/U is used to locate faults on telephone and telegraph equipment

6.2 **Ordering data.** — Procurement documents should specify the following

(a) Title, number, and date of this specification and any amendment thereto

(b) Type of packaging and packing required for shipment (immediate use at first domestic receiving activity, domestic shipment and indefinite storage, or overseas shipment).

(c) Samples required

(d) Marking and shipping of samples

(e) Place of final inspection

6.3 **Nomenclature.** — The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example AN/PIQ-1W As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information.

6.4 **Group C inspection.**—Approval to ship may be withheld, at the discretion of the



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Government inspector, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.3.2.2.)

Notice —When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy—Bureau of Yards & Docks
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